INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 5: Discussion

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5 Discussion

Chapter 4 reported the measurements on Arbitrum One. Here each research question and objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. It is the main analysis, and none of its results is a registered result (Section 3.5.3). Unless noted otherwise, same-window statements concern the matched cohort of contracts and out-of-window statements the spender cohorts.

5.1 Findings by research question

5.1.1 RQ1: The two single-layer walks and their cost

RQ1 asks whether EndorseRank and AWP order the same contracts differently, whether each agrees most with current checks of its own edge type, and what each walk, the coupled one included, costs. They order the contracts differently: the interval of their agreement excludes both zero and one, whether a contract missing from a graph is scored zero or kept as an isolated node (Table 4.6). The coefficient compares fewer contracts than the cohort holds: four in five cohort contracts grant or receive no capped allowance between contracts and tie at zero under EndorseRank (Table 4.2), so both walks order only about a third of the contract pairs, and about a quarter of those in opposite directions (Section 4.4).

The divergence follows from how the graphs are built. The allowance graph records the latest

allowances between contracts that the owner's balance backs and is far smaller than the transfer graph, which records every transfer (T. Do et al., 2023), and the two methods restart differently (Table 4.3). H1 is supported, largely by construction. One score cannot stand in for the other, and the divergence is also the premise of C-PR: if the two walks ordered the contracts alike, one walk over both could add little to either.

In the same window each score follows the edges it walks on: the walks over transfers agree most with the transfer family, the walks over allowances with the allowance family, and every interval on a score's own family excludes zero (Table 4.5). Circularity bounds H2: every proxy is computed from the same events between contracts as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955) and not an external test. The rule for missing contracts also moves EndorseRank's coefficients: as isolated nodes, the contracts outside the allowance graph share the score of a node without in-edges, which raises its agreement with the allowance family and lowers it with the other two (Table 4.6). In the narrow sense of agreement with the checks of its own edges, H2 is supported, largely by construction (paired differences in Appendix 8.5).

The cost of each walk follows the size of its graph. EndorseRank is the cheapest score in time, memory and iterations; C-PR, on the union of both layers, takes about one tenth longer than AWP, and S-PR about as long as AWP (Table 4.3). AWP's time ratio to EndorseRank lies above the edge ratio at $d = 0.85$ and 0.95 and below it at $d = 0.75$ (Table 4.7), so the cost grows with the size of each graph (Langville & Meyer, 2006) and with the iterations its walk needs, which the benchmark does not separate; the speed belongs to the smaller graph, not to the algorithm. The scaling table qualifies this: its subsample rows keep only the pairs incident to the sampled contracts and its last row is the whole contract graph, so the rows are not sizes of one growing graph (Table 4.4). The damping and value-slope checks leave the alignment pattern as it was (Tables 4.7 and 4.8). H3 is supported, again largely by construction. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ2–RQ5.

5.1.2 RQ2: Discovery order

RQ2 asks whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than breadth-first and backlink-count crawls. The discovery-order test was defined after the registered results were known and decides no rule of the plan or the registration. By the decision rule of its protocol (Section 3.7), the answer is yes only if, on each importance, the intervals of D_1 (C-PR′ minus breadth-first) and of D_2 (C-PR′ minus the backlink count) lie above zero at T_{obs} in the primary reveal mode, in which a visit reveals a contract's in-edges and out-edges. On both importances the interval of D_1 lies above zero and that of D_2 below zero (Table 4.12), and the verdict reads: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The answer to RQ2 is no.

C-PR′ is sooner than breadth-first order on both importances, but on user-authorized value the margin is small and does not hold in every sensitivity, and in no row does C-PR′ lead the backlink count (Table 8.9). The one single-layer walk that does better than its count is AWP′: it reaches the hot contracts of both importances sooner than the transfer backlink count, has the highest $\text{AUC}_{10\%}$ of the compared crawls and leads C-PR′ on both importances, whereas C-PR's transfer layer walked alone, the same edges with other weights and restarts, lies below that count (Tables 4.13 and 4.14). Every crawl stays far from the ideal order, and under every ordering most of the first tenth of the visits goes to contracts with no user-account demand (Table 8.10).

Two facts bound this reading. First, the result depends on what a visit reveals. In the `out` mode, which reveals out-edges only and whose universe is the giant strongly connected component with the contracts reachable from it, all four contrasts lie above zero at both freezes; at t_1 in the primary mode they repeat the signs of the decision (Table 4.12). The `out` mode carries no verdict: the

protocol makes the mode on the weakly connected component primary when the other universe holds fewer than half of its contracts, as the structure pilot found (Table 8.7). The modes differ in the universe, in what the crawl sees of a frontier contract and in how often C-PR′ picks by a tie (Table 8.10); which difference reverses the comparison with the backlink count was not tested. Second, the allowance layer between contracts is thin: a capped allowance in-edge reaches about half a percent of the contracts of SCC ∪ OUT at t_1 in the structure pilot (Section 3.7). In the deciding run C-PR′ is therefore close to a transfer walk: its allowance layer raises the area by a small, resolved amount on both importances (Table 4.14), and in the `out` mode at T_{obs} it lowers the area on user-account inflow (Appendix 8.4.4).

5.1.3 RQ3: Out-of-window prediction and the raw counts

RQ3 has two clauses. The first asks whether frozen scores predict the new owner–spender approval pairs and the new transfer senders that the spender contracts gain in the following three months. Every score predicts both labels in both windows, and the edge type decides how well (Tables 4.15, 4.20 and 4.18). On new approval pairs the in-approve degree leads and the scores built on transfers alone rank lowest; on new senders AWP and the transfer in-degree lead and the scores built on allowances alone rank lowest. Two facts frame every coefficient: more than four in five of the new pairs and senders come from user accounts, which no score reads (Table 8.6), and about 27% of each spender cohort lies outside the union graph and scores zero (Table 4.2).

The second clause asks whether any walk beats the raw count of the edges it smooths; the answer differs by walk. EndorseRank ranks new approval pairs below the in-approve degree in both windows, with either restart rule (Tables 4.16 and 4.18; Section 4.5.4). AWP ranks new senders above the transfer in-degree in both windows, with intervals just above zero, but among the W0 spenders that another contract had paid before the freeze, which the count can order, it ranks them below the count (Table 4.17), a restriction chosen after the results were known and not computed for this contrast in W1. C-PR ranks new approval pairs above the union in-degree, the count of the edges it walks, and new senders below it, with larger gaps inside the union graph (Table 4.23); it lies be-

low the in-approve degree on new approval pairs and below the transfer in-degree on new senders (Table 4.22). So of the single-layer walks AWP alone gains on its count, by one or two hundredths of τ on the whole cohorts and inside the union graph, and C-PR gains on the count of its own edges on one label and loses on the other. Known results lead one to expect the count to be hard to beat: in growing networks new links attach preferentially to nodes that already have many (Barabási & Albert, 1999), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022), and where degree correlations are weak PageRank is on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is a reweighted count of the contracts that authorize them (Section 5.3.2). All of these contrasts were computed after the labels were known.

5.1.4 RQ4: The coupled walk against its layers

RQ4 and RQ5 concern the method of the thesis, and for C-PR, the coupling the plan fixed, both get a negative answer under their own rules, applied to the scores of the contract graph as a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6). RQ4 asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan, it does not. On W0 C-PR ranks new approval pairs below its allowance layer and new senders below its transfer layer, so both conditions of the primary part fail (Table 4.16), and in the same window the secondary part fails on all three families (Section 4.8.1). The plan fixed the rule and λ before the study extracted any log, so λ could not have been tuned to it. The result limits an obvious next step: a convex mixture of the two edge types at the transition level ends between its layers on both labels (Table 4.22).

Two results outside the rule qualify the loss to the allowance layer. Three in four spenders score zero under that layer, and among the spenders inside the union graph the contrast of C-PR with it on new approval pairs does not exclude zero in either window (Tables 4.2 and 4.23). And when every positive allowance counts at its own value, the allowance layer walked alone ranks new approval pairs much lower and C-PR lies above it in W0 (Table 4.11). Neither result decides the rule, and

the loss to the transfer layer on new senders stays in both.

The ablation and the layer weights explain part of this answer: S-PR, which restarts the transfer walk from the allowance walk, ranks new senders level with the transfer layer and new approval pairs far below the allowance layer (Table 4.16), and C-PR ranks new approval pairs above it in both windows, so the allowance signal survives better where allowance edges carry part of the walk than where they only seed its restarts. The layers of C-PR carry dollar amounts and restart uniformly, whereas AWP bounds the weight of each transfer and restarts at active senders; walked alone, C-PR's transfer layer already ranks new senders below AWP, without a paired contrast (Tables 4.15 and 4.20), so part of C-PR's gap to AWP comes from the weights and restarts, not from the coupling.

5.1.5 RQ5: Allowance edges added to a transfer walk

RQ5 asks whether adding the allowance layer improves a transfer walk on new data, gaining on later approvals without losing on later inflow. Rule B was registered with W1 before the study extracted the July–September logs. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. On W1 F1 holds, as the same contrast did on W0 (Table 4.22): C-PR ranks new approval pairs above its transfer layer walked alone. F2 fails: on new senders C-PR lies below that layer by more than the registered margin of 0.01, with its whole interval beyond the margin (Table 4.21). Rule B is not met. The two registered sensitivity analyses give the same reading, and so do the spenders inside the union graph and the uncapped allowance layer, which decide no rule (Tables 4.23 and 4.11).

Beside the rules, the contrasts of Table 4.22 set C-PR against every other score and show losses on both labels: on new approval pairs C-PR lies below EndorseRank with either restart rule, its own allowance layer and the in-approve degree and value in both windows, and on new senders below AWP and the transfer and union in-degrees, and level with or slightly below S-PR. Among the spenders that another contract had paid before the freeze, C-PR ranks new approval pairs above the in-approve degree in both windows, and its lead over AWP there is smaller in W0 and not resolved in W1. What the allowance layer adds to a transfer walk out of window is therefore information

about later approvals, bought with a small, resolved loss on later inflow, and a walk or a count over allowances alone carries that information better. These contrasts were computed after the labels were known and decide nothing.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

5.3 Theoretical implications

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999), and Cho et al. (1998) found that a crawler ordering the pages it has seen by PageRank reaches important pages sooner than one ordering them by backlink count or breadth-first (Section 2.6). H.-D. Do and Do (2023) carried the citation reading over to Ethereum transactions, and AWP (T. Do et al., 2023) added value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner that lets a spender act for it issues an endorsement of spending authority (Vogelsteller & Buterin, 2015), and C-PR lets one walk follow both relations. The RQ1 divergence shows that changing what the edges mean, and where the walk restarts, changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each valid relative to the relationships its edges encode.

For C-PR that result does not carry over to contracts in the deciding run: on the hot set of the contract backlink count, the comparison that repeats the one of Cho et al. (1998), the count ordering is far sooner than C-PR′, and on user-account demand it is sooner as well (Table 4.14). The out-of-window results add a qualification that the propagation metaphor hides: the edge type decided which label a score could anticipate, but of the single-layer walks only AWP added to the count of its own edges, and not among the spenders that another contract had paid (Section 5.1.3). On these

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of the τ between EndorseRank and AWP contains neither 0 nor 1; a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of a score's own edges is an intended-construct check; the runtime ratio is given alongside the edge ratio, under a single fixed protocol	Table 4.6, last row: the interval lies strictly inside $(0, 1)$. Table 4.5: every walk over transfers agrees with the transfer family and both walks over allowances alone with the allowance family, each interval excluding 0. Tables 4.3 and 4.7: time ratio beside edge ratio at each damping value	Met (largely by construction)
O2	Decision rule of the discovery-order protocol, read at T_{obs} in the primary mode: on each importance the verdict is V1 when the 95% intervals of D_1 (C-PR′ minus BFS) and D_2 (C-PR′ minus IB'_U) both lie above 0, V2 when only that of D_1 does and V3 otherwise; the answer is the weaker of the two verdicts	Table 4.12: on both importances the intervals of D_1 lie above 0 and those of D_2 below 0. Verdict V2 on both	Not met
O3	A score predicts a label when its interval excludes 0; a walk beats a count only when the interval of the paired difference lies above 0	Tables 4.15, 4.20 and 4.18: every score predicts both labels in W0 and W1. Tables 4.16, 4.17 and 4.18: AWP lies above the transfer in-degree on the whole cohorts and below it among the W0 spenders another contract had paid; EndorseRank lies below the in-approve degree with either restart rule. Table 4.22: C-PR lies above the union in-degree on new approval pairs and below it on new senders. All computed after the labels were known	First clause met; second met by AWP on the whole cohorts only, and by C-PR against the union in-degree on new approval pairs only
O4	Rule A of the analysis plan: on W0, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the leading layer's interval on the transfer and allowance families, and it exceeds both layers on the stability family	Scores of the contract graph, re-analysis. Table 4.16: both conditions of the primary part fail. Same window: C-PR's point estimates lie outside the leading layer's interval on both families (Table 4.5), and on the stability family it lies between the layers (Table 4.19); see Sections 4.8.1 and 4.8.2	Not met, as in the registered analysis (Table 8.16)
O5	Two of the registered criteria of rule B (registration of W1): on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs (F1) and is not worse by more than 0.01 on new senders (F2: lower 95% bound above -0.01). Under the registration the rule holds only if each of its criteria holds	Scores of the contract graph, re-analysis. Table 4.21: F1 holds; the interval of F2 lies wholly below -0.01, so F2 fails. The two registered sensitivity analyses read the same	Rule B is not met; F1 held and F2 failed in the registered analysis as well (Table 8.16)

data the choice of edge decides which contracts a score can order, and a count of that edge orders them about as well as a walk over it or better.

5.3.2 Depth-one spenders: what a walk adds to a count

The allowance graph keeps the latest allowances between contracts that stay positive under the balance cap (Section 3.6). Most of its nodes are pure owners, with out-edges only, or pure spenders, with in-edges only, but a contract can be both, so the graph also has paths of more than one step. For a spender whose owners receive no allowances, a walk with uniform restarts has a closed form.

Proposition. Let PageRank with transition matrix P be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u P_{uv}, \tag{5.1}$$

the sum of the transition probabilities into v. In EndorseRank $P_{uv} = w(u,v)/o(u)$, so $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all the spenders it approves. In C-PR P_{uv} is the coupled transition of Equation (3.9), and an in-neighbor without in-edges is one that has none in either layer.

Derivation. At the fixed point, $r_v = (1 - d + d\,r_D)/|V| + d\sum_u r_u\,P_{uv}$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1-d+d\,r_D)/|V|$. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On EndorseRank's graph at T_{obs}, 651 of the 926 cohort contracts that are spenders have depth one, and EndorseRank lies within 1% of Equation (5.1) for each of them; of the deeper spenders, which have an owner that itself receives allowances, 33.8% do. Over all 926 the share is 80.3%, and over the 1,362 W0 spenders of the graph at t_1 it is 68.4%. Over the cohort spenders EndorseRank's Kendall τ with δ is 0.923, a point estimate. These shares were not computed for the union graph

of C-PR.

Two consequences follow. First, for a depth-one spender EndorseRank only reweights the count of the contracts that authorize it, counting an owner that authorizes many spenders for less at each; besides the time and value weights, that normalization is all that separates it from the count, which ranks new approval pairs better (Section 5.1.3). For these spenders EndorseRank's negative answer to the second clause of RQ3 thus has an analytic basis: there was no propagated information for the walk to add, and with activity restarts the variant is likewise a one-step sum over the owners, weighted by their restart mass. Second, for most deeper spenders the score departs from the closed form, so multi-hop propagation is present in this graph, but over all spenders EndorseRank did not beat the raw count out of window, and the deeper spenders were not tested apart. The edges are read from the whole chain, so the short paths are no artifact of a sample. The one-step form reappears in the discovery-order test: the diagnostic δ', the normalized in-strength on the visited part with C-PR's mixed weights, has a higher $\mathrm{AUC}_{10\%}$ than every walk and every count ordering in all four runs (Table 4.13); it stands outside the comparisons of the protocol and has no paired contrast.

5.3.3 Multi-construct domain alignment

Each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs (Cronbach & Meehl, 1955), and the integrated view of Messick (1995) goes further, counting the consequences of using a score as part of its validity. The same theory says what such a pattern cannot settle: the instrument that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. The importance measures of the discovery-order test come closest to a criterion that no instrument was built for, since no ordering reads an edge of a user account; they are still sums of transfers and allowances, so they test whether a position among contracts agrees with demand from users, not whether either makes a contract worth relying on.

5.4 Practical implications and deployment scenarios

A wallet interface, a block explorer, an auditor or an integrator faces very many contracts and has to choose which to look at first. The results do not single out one ordering for that choice: in the deciding discovery-order run a crawl ordered by the contract backlink count reached the contracts with the most user-account demand sooner than C-PR′, and out of window no walk clearly improved on the leading count of a label.

5.4.1 The coupled walk and combined use

Folding the two edge types into one walk did not pay against the published transfer walk on these data: C-PR costs slightly more than AWP (Table 4.3), trails it on new senders in both windows and on both importances in the deciding discovery-order run, and leads it out of window only on new approval pairs (Sections 5.1.2 and 5.1.5). An operator free to choose should keep the signals apart, using a count or a walk over allowances where authorization matters and AWP or the transfer count where flow matters. For an operator tied to a single transfer walk weighted by dollar amounts, adding allowance edges bought information about later approvals that held on new data, at a loss on inflow larger than the registered margin; rule B is not met.

5.4.2 Transfer-hub and flow monitoring

A protocol that describes current flow can read it from in-degree and in-value directly. AWP (T. Do et al., 2023) had the highest coefficient of the walks on later new senders (Tables 4.15 and 4.20) and ordered the crawl with the highest area in the deciding discovery-order run (Table 4.13), the best record of the walks in this study. Its margin over the transfer in-degree out of window is about a hundredth of τ and reverses in W0 among the spenders that the count can order (Table 4.17), so little is lost by counting.

5.4.3 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most, there is reason to use allowance edges. The cheapest form is the spender's in-approve degree among contracts, which needs no solver and ranked new approval pairs above EndorseRank, with either restart rule, and above C-PR in both windows (Tables 4.16, 4.18 and 4.22). EndorseRank, its propagated version, solves in a fraction of AWP's time (Table 4.3), but for most spenders it reflects the authorizing contracts only through how many spenders each authorizes (Section 5.3.2). Any allowance score between contracts covers little: one cohort contract in five and one spender in four grant or receive one (Table 4.2). And the balance cap matters: when every positive allowance counts at its own value, the in-approve degree and EndorseRank have about half the coefficient on new approval pairs (Table 4.11; point estimates), so a tool that reads allowances to anticipate authorization should count each one up to the balance behind it.

5.4.4 Industry adoption scenarios

- Wallet interfaces and security tooling. A user asked to approve a spender wants to know whether it is widely authorized. A badge that shows how many contracts with a balance have authorized a spender, or its EndorseRank, makes that delegation explicit (Lundfall, 2020; Vogelsteller & Buterin, 2015); it conveys the position of a graph node and is neither a security audit nor a guarantee against malicious contracts, and it is empty for the spenders, at least three in four, that no contract with a balance authorizes.
- Block explorers, auditors and integrators. A service that labels or indexes contracts, a reviewer choosing which contracts to read first and an aggregator choosing what to integrate cannot treat millions of contracts alike and need an order of work. In the deciding run the contract backlink count, which needs no solver, reached the contracts with the most user-account demand sooner than C-PR′, and most early visits of every ordering went to contracts without such demand (Section 5.1.2). An ordering by position among contracts is a first filter at best

and says nothing about the quality of a contract's code.

- Lending protocols. A ranking of the contracts that users rely on could later serve as one input to credit decisions about contracts, for instance when a protocol weighs lending to a vault or a strategy contract. This thesis does not test that use: no score was compared with a default, repayment or loss outcome.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each can recompute it. C-PR, EndorseRank and AWP can be recomputed with the open pipeline from public logs, public prices and the code at each address (Appendix 8.1).

5.5 Societal and practical impact

On a rollup with millions of contracts, finding out which of them others rely on is expensive, and the capacity to do it is unevenly spread: wealthy actors may afford proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. An ordering built from public transfers and allowances between contracts, cheap to recompute (Table 4.3), lowers the barrier to a first look at the contracts that ask for permissions. An allowance that stays in force is one route by which tokens are drained (Section 2.18), but no score is shown to anticipate drains: on W0 the drain heuristic and the revocations rose with every score, the transfer-based ones included, so they track volume and are not offered as evidence (Section 4.7).

A score built from public events and public prices with versioned parameters supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed (Zetzsche et al., 2020), and it brings the duties of Section 5.6.

5.6 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Legal scholarship has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024) warn that decentralized scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. C-PR, EndorseRank and AWP are computed deterministically from public data with open-source algorithms, so anyone can recalculate them (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with little activity, new ones among them, will score lower whatever their code, and a contract that only user accounts pay or authorize has no position in the graph.

Ethically, these scores should never be used as security ratings or as measures of the ability to repay; they are rank-based indicators of a contract's position in the defined graph. A deployment that grants or withholds a service by score thresholds should document the graph definitions behind the scores, the timeframe and the limitations.

Privacy on an open blockchain is limited by the ledger's very nature (Nakamoto, 2008), and the scores use public data only. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study scores contract addresses alone and makes no attempt to link any address to a real-world individual (Section 3.17).

5.7 Threats to validity

5.7.1 Construct validity

The threats below concern whether the scores and checks measure what they are read as measuring.

- The scores are not ratings of default risk: no default, repayment or loss label enters the anal-

ysis.

- Same-window checks rest on local allowance and transfer statistics between contracts, and own-construct agreement is partly mechanical: EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for incoming transfers weighted by their value (Table 8.11); EndorseRank's coefficients also depend on the rule for missing contracts (Table 4.6).
- Importance in the discovery-order test. User-account inflow and user-authorized value are measured on-chain, on five tokens, from user accounts only. They record dollars that user accounts moved or authorized, not whether a contract is needed, safe or valuable, and no label of importance from outside the chain was used. Beside the 16.61 billion dollars of inflow to typed contracts at T_{obs}, user accounts sent 1.68 billion to addresses that were not typed because only user accounts transact with them (Section 3.15); a contract among these has no edge to another contract and lies outside every crawl. Transfer volume can be inflated by wash trading (Victor & Weintraud, 2021); the cap per account bounds one account and not one actor.
- Out-of-window construct. A later approval pair is still an endorsement event and a later sender a flow event, so neither is an independent construct, and as counts of arrivals they favor any score that tracks the current count of counterparties (Section 5.1.3). They count mostly user accounts, which no score reads (Table 8.6), and the part that comes from contracts was not evaluated on its own.
- Account types. Every node is an address typed as a contract by its code at one block, read after both label windows (Section 3.5.1); a contract that destroyed itself before EIP-6780 (Ballet et al., 2023) is classed as a user account. Safes and ERC-4337 accounts are user accounts by rule and by listed code families, and kinds of smart account that the rule misses stay classed as contracts: after the lists, $115{,}084$ typed contracts still carry a `validateUserOp` selector and no paymaster selector, and the contracts that services create for each user, order or payment

are not picked out (Appendix 8.4.3). Users may control such contracts, so some edges of the graph are in effect edges of user accounts.

- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with a slope set so that the median amount weighs one half (Section 3.5.3), and the slopes were checked only in the same window (Table 4.8). Under the balance cap the median positive allowance is 0.071 dollars, so V is at its ceiling for any allowance backed by a dollar or more and separates very small balances from the rest and little else.

5.7.2 Internal validity

The threats below concern whether the comparisons are free of choices that could have produced their results.

- Shared score windows. Every method and proxy is computed over the twelve months before its freeze, and market regimes peculiar to those months may push alignment up or down. An allowance last set before the window began is not an edge even while it is in force, and an allowance is an edge only if its owner holds a balance at the freeze, so the allowance layer is a snapshot of one instant, and its size differs severalfold between the two freezes (Table 4.1). No check varies the calendar window.
- One owner contract at T_{obs}. One owner contract holds 85% of the capped allowance pairs at T_{obs}; at t_1 the largest owner holds 256 pairs (Table 4.1). The same-window results on the allowance side and the W1 scores are computed on that layer, so one contract's allowances weigh on all of them; most of its pairs lie outside the universe of the discovery-order test (Section 4.6), and the thesis does not identify the contract. W0, scored at t_1, has no such owner, and every contrast of Table 4.22 has the same sign in W0 as in W1.
- The thin allowance layer. Between contracts, capped allowances are few beside transfers: about one pair for every twenty-two transfer pairs at T_{obs} and far fewer at t_1 (Table 4.1). The coupled walk is therefore close to its transfer layer, and every comparison of C-PR with that

layer, in the discovery-order test above all, measures a small perturbation.

- Cohort selection and coverage. The matched cohort is the $6{,}443$ contracts that received non-zero approvals of one of the five tokens from at least three owners in the twelve months before T_{obs}; the rule counts every approving account, and results may not carry over to other contracts. About a third of the matched cohort and 27% of each spender cohort end no pair between contracts and score zero under every score and baseline (Table 4.2), so on the whole cohorts a contrast also reflects how two scores order the members with a pair against those without one. Among the members inside the union graph the main contrasts keep their direction, except that of C-PR with its allowance layer on new approval pairs, which does not exclude zero there (Table 4.23); the sample-definition stages show how the coefficients move with the evaluation population (Table 4.9).
- Reveal mode of the discovery-order test. The decision is read in the mode the protocol made primary, and in the other mode D_2 has the opposite sign at both freezes (Section 5.1.2); the test does not say which mode is closer to how its users meet contracts. Its intervals resample the hot contracts for twenty fixed seeds, and under a bootstrap over the seeds D_2 on user-account inflow and D_1 on user-authorized value are not resolved (Table 8.9). The two freezes share nine months of their score windows and are not independent.
- Order of analysis. The cohort rule, scores, labels, statistics and rule A were fixed in the analysis plan before any log of the study was extracted, and rule B with its margin and its two sensitivity analyses was registered after the W0 results were known and before the study extracted the July–September logs (Section 3.8.1). The main analysis, with the definitions listed at the start of this chapter, was defined after the registered results were known, so it is post hoc: its verdicts under rule A and under F1 and F2 are a re-analysis, not registered tests, and the labels of both windows were known when it was defined. The registered analysis reached the same verdicts, although some contrasts differ, among them the sign of C-PR against its allowance layer on new approval pairs in W0, which lay above zero there (Table 8.16).

- Comparators and baseline fidelity. The rules compare C-PR with its own layers walked alone, the only contrast that changes the coupling and nothing else; its contrasts with EndorseRank and AWP mix the coupling with their weights and restarts (Table 4.22). AWP follows the published method in its edge weights and restarts (Section 3.6.1), but the paper gives no parameter values, so k, t_0 and b are this study's choices, and here a sender's activeness counts only its transfers of five tokens to contracts.
- Dependence, balances and timing. Contracts that share owners or a protocol are resampled as independent, and the post hoc contrasts are not adjusted for multiplicity (Section 3.11); the five tokens were chosen by their rows among the events of the plan's cohort up to T_{obs}, which include the label window of W0. Balances are reconstructed from `Transfer` logs: in a check, 99.5% of $1{,}500$ sampled balances equaled `balanceOf` at the block of T_{obs} (Section 3.5.2). Runtimes are wall-clock times from one machine (Appendix 8.1).

5.7.3 External validity

The threats below concern how far the results carry beyond the data studied.

- Single chain and window. The evidence comes from Arbitrum One alone, from contracts that at least three owners approved on one of five tokens, over the twelve months before each of two freezes three months apart and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or later months would need the same tests there, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).
- Five tokens and edges between contracts. A transfer or an allowance of another token, a payment in native ether and any event with a user account at one end are not edges, so a contract that users rely on directly, without another contract paying or authorizing it on these tokens, has no position in the graph (Table 8.8).
- Path depth. Few contracts both grant and receive a capped allowance, so the findings about propagation on the allowance layer do not carry over to graphs with longer endorsement paths

(Section 5.3.2).

5.8 Limitations

On this sample, the strengths of C-PR are a crawl that reaches the contracts with the most user-account demand sooner than breadth-first order, a gain from its allowance layer on later approval pairs and, by a small amount, in that crawl, and a ranking of later approval pairs above AWP and of later senders above every allowance-side score. Its weaknesses are a crawl that is no sooner than the backlink count and later than AWP's, the failure of rule A and of F2, rankings of later senders below AWP and the transfer and union in-degrees and of later approval pairs below EndorseRank, its own allowance layer and the in-approve degree, a stability-family coefficient below AWP's and the highest cost of the walks.

The coupled operator was tried at three fixed values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance counted up to its owner's balance and up to U. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors in place of the transition rows, and other Markov chains are not evaluated, so the answers to RQ2, RQ4 and RQ5 apply only to that family of methods on this dataset. EndorseRank, the allowance reference, orders one cohort contract in five and ranks later approval pairs below the raw in-approve degree. None of the scores says anything about contracts that few owners approve or about user accounts, which are not scored (Section 3.2).

The data add four limits. The graphs hold five tokens only (Table 8.3), so a contract that owners approve only on other tokens is outside the cohort. Every price comes from one public service, the daily series of DefiLlama (DefiLlama, n.d.), so an error in it on some date would pass into every amount of that day (Appendix 8.4.2). The account type was read once, from code at one block, and the detection of smart accounts is incomplete (Table 8.4). And importance is measured on-chain from user accounts only (Section 5.7.1).

None of the scores uses off-chain information about a contract; their only off-chain inputs are the token list and the daily price series.

For this cohort of contracts and time frame, neither C-PR nor EndorseRank is a rating of default risk or a security audit. On these data counts were hard to beat: the backlink count in discovery and the in-approve degree on later approvals, with AWP the one single-layer walk that added to its count. Chapter 6 restates what has been shown, what that knowledge is worth and what it would take to extend these bounds.